Uma R. Mohan, Ph.D.

CONTACT:

National Institutes of Health
Functional Neurosurgery Section
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Github: https://github.com/umarmohan/

EDUCATION:

Columbia University New York, NY
PhD in Biomedical Engineering August 2021
Human Electrophysiology, Memory, and Navigation Lab
Dissertation: *Characterization and modulation of neural signals that support human memory*

Cornell University Ithaca, NY
M.Eng. in Biomedical Engineering, Focus in Bioinstrumentation May 2014

The Johns Hopkins University Baltimore, MD
B.S. in Biomedical Engineering, Cell and Tissue Engineering focus May 2013
B.S. in Applied Mathematics & Statistics

RESEARCH EXPERIENCE:

National Institutes of Health Bethesda, MD
National Institute of Neurological Disorders and Stroke February 2022-present
Postdoctoral IRTA Research Fellow
Advisor: Kareem Zaghloul, MD, PhD

- Targeting open-loop and closed-loop stimulation to evoke and disrupt hippocampal ripple activity, respectively, to modulate memory performance.
- Characterizing, modeling, and predicting the effects to different types of direct brain stimulation on large-scale neural activity and single-unit neuronal activity using a control-theoretic framework
- Analyzing local directional patterns of cortical traveling waves measured from microelectrode Utah arrays in associative memory encoding and retrieval
- Determining the role of large-scale cortical traveling waves in cued attention

Columbia University: Department of Biomedical Engineering New York, NY
Postdoctoral Researcher September 2021-February 2022
Advisor: Joshua Jacobs, PhD

- Characterized the relationship between spatiotemporal patterns of intracranially measured neural oscillations called traveling waves and memory encoding and retrieval behavior

Columbia University: Department of Biomedical Engineering New York, NY
Graduate Researcher January 2016-August 2021
Advisor: Joshua Jacobs, PhD

- Characterized how direct electrical stimulation of the human cortex at different frequencies, amplitudes, and locations affect neuronal activity to support optimization of stimulation protocols designed to modulate memory
- Developed methods to detect and reject stimulation artifact from direct electrical brain stimulation as part of the DARPA-Restoring Active Memory (RAM) project
- Studied direct brain recordings from epilepsy patients to examine how brain stimulation differentially affects a variety of neuronal activities during spatial and verbal memory tasks

Advisor: Paul Sajda, PhD (semester rotation) August 2015-January 2016

- Investigated link between scalp EEG patterns and pupil diameter during tactile oddball paradigm

Johns Hopkins University: Department of Neuroscience
Baltimore, MD
Computational Neuroscience: Undergraduate Researcher May 2012-May 2013
Advisor: Ernst Niebur, PhD

- Designed models of neural activity across regions during visual tasks problem sovling as part of IARPA-Integrated Cognitive-neuroscience Architectures for Understanding Sense-making (ICArUS) project
- Simulated interactions between brain regions during executive control and attention.

Johns Hopkins University: Department of Biomedical Engineering Baltimore, MD
Spinal Cord Injury: Undergraduate Researcher April 2010-February 2012
Advisor: Angelo All, PhD

- Researched the benefits of therapeutic hypothermia following spinal cord injury using somatosensory evoked potentials, motor evoked potentials, histological results, and functional outcomes.
- Investigated unmyelinated motor system of the leech model for generation of myelin in the CNS.

Johns Hopkins University: Departments of Biomedical Engineering, Pathology Baltimore, MD
Center for Bioengineering and Design and GlaxoSmithKline August 2011-May 2013
Advisor: Anne Le, MD

- Designed and prototyped affordable, robust point-of-care device to screen for G6PD enzyme deficiency.

Case Western Reserve University: Department of Neurosciences Cleveland, OH
Developmental Neurobiology: Student Summer Researcher June 2006-September 2008
Advisor: Heather Broihier, PhD

- Investigated role of extracellular signals in shaping activity of transcription factors for neuronal communication.

GRANT FUNDING:

NINDS Competitive Fellowship Award (NCFA) for the “Characterization and prediction of the effects of direct electrical stimulation on neuronal activity in humans.”

NINDS Intramural equivalent of F32

INDUSTRY EXPERIENCE:

Cardiopulmonary Corp. Milford, CT
Software Engineer June 2014-August 2015
- Designed software platform for hospital benchmarking of alarm management
- Incorporated data integration methods, reporting software, and statistical analytics.

West Pharmaceutical Services Exton, PA
Pharmaceutical Drug Delivery Systems: Internship partnered with Cornell June 2013-May 2014
- Designed novel mechanism and constructed functional prototype for delivery of high viscosity drugs from self-administration injection devices.

Food and Drug Administration: Center for Devices and Radiological Health Silver Spring, MD
Mathematical Modeling of Anesthetic Drug Delivery Devices: Summer Intern June-Sept 2012
- Designed mathematical simulations for autonomous anesthetic drug delivery devices by integrating PK/PD models and reinforcement learning to map drug infusion rates to physiological parameters.

PEER REVIEWED PUBLICATIONS:

Mohan, U.R., Jacobs, J. Why does invasive brain stimulation sometimes improve memory and sometimes impair it? *PLOS Biology* 22(10): e3002894 (2024). Link

Mohan, U.R., Zhang, H., Ermentrout, B., Jacobs, J. The direction of theta and alpha travelling waves modulates human memory processing. *Nat Hum Behav* **8**, 1124–1135 (2024). Link

- Featured hero image on journal website

Qasim, S.E., **Mohan, U.R.,** Stein, J.M., Jacobs, J. Neuronal activity in the human amygdala and hippocampus enhances emotional memory encoding. *Nat Hum Behav* **7**, 754–764 (2023). Link

Lorusso N.D.*, **Mohan, U.R.*,** Jacobs, J. Jose Delgado: A controversial trailblazer in neuromodulation. *Artificial Organs*. 46:531–540 (2022). Link
*equal contributions

Mohan, U.R., Watrous, A.J., Miller, J.F., Lega, B., Sperling, M., Worrell, G., Gross, R., Zaghloul, K.A., Jobst, B., Davis, K., Sheth, S., Stein, J., Das, S., Gorniak, Wanda, P.A., R., Rizzuto, D., Kahana, M., Jacobs, J. The effects of direct brain stimulation in humans depend on frequency, amplitude, and white-matter proximity. *Brain Stimulation*. 13, 1183–1195 (2020). Link

MANUSCRIPTS INVITED:

Mohan, U.R., Jacobs, J. Why does invasive brain stimulation sometimes improve memory and sometimes impair it? PLOS Biology 22(10): e3002894 (2024). Link

TALKS & INVITED SEMINARS:

Mohan, U.R. Cortical traveling wave pattern reinstatement supports successful associative memory formation and retrieval in humans. Cognitive Neuroscience Society. March 6-10, 2026. *Data Blitz Talk*

Mohan, U.R. Targeting memory-related neural oscillations with direct electrical stimulation. Albert Einstein College of Medicine, Leo M. Davidoff Department of Neurological Surgery Seminar. March 2, 2026.

Mohan, U.R. The direction of theta and alpha traveling waves modulates human memory processing. The 48th Annual Meeting of the Japan Neuroscience Society. Symposium Title: "[1S06m] Understanding Human Cognition through Intracranial Neurophysiology: Bridging Basic and Clinical Research". July 24–27, 2025.

Mohan, U.R., The direction of theta and alpha traveling waves modulates human memory processing. Center for Neuroscience Imaging Research, Institute for Basic Science, Sungkyunkwan University, ABC Seminar Series. June 26, 2025.

Mohan, U.R. Direct electrical microstimulation of the hippocampus evokes high frequency ripple oscillations in humans. Spring Hippocampal Research Conference. May 18-23rd, 2025. *Data Blitz Talk*

Mohan, U.R. Direct electrical microstimulation evokes high-frequency ripple oscillations in humans. NIH Surgical Neurology Branch Scientific Lecture Series. March 6th, 2025

Mohan, U.R., Jacobs, J, Zaghloul, K.A.**.** The modulation of spatiotemporal patterns of neural activity depend on the frequency of direct electrical stimulation in humans**.** 2025 6th International Brain Stimulation Conference. February 23-26, 2025. *Symposium talk.*

Mohan, U.R., Zhang, H., Ermentrout, B., Jacobs, J. The direction of theta and alpha traveling waves modulates human memory processing. Society for Neuroscience. Oct. 5-9, 2024. *Nanosymposium talk.*

Mohan, U.R., Zhang, H., Ermentrout, B., Jacobs, J. Characterization and modulation of memory-related traveling waves in humans. UVA Cognitive Area Brown Bag Speaker Series. Sept, 26, 2023. *Invited talk*.

Mohan, U.R., Zhang, H., Ermentrout, B., Jacobs, J. The direction of theta and alpha traveling waves modulates human memory processing. 2023 International Conference on Learning and Memory. April 26-30, 2023. *Lightning talk*.

Mohan, U.R., Zhang, H., Jacobs, J**.** Direct cortical stimulation modulates propagation directions of traveling wave in humans**.** 2023 5th International Brain Stimulation Conference. February 19-22, 2023. *Talk and symposium chair.*

Mohan, U.R., Zhang, H., Ermentrout, B., Jacobs, J. The direction of theta and alpha traveling waves modulates human memory processing. Dugué Lab Seminar. February 1, 2023. *Invited talk*.

Qasim, S.E., **Mohan, U.R.**, Stein, J., Jacobs, J. Neuronal activity in the human amygdala and hippocampus enhances emotional memory encoding. Society for Neuroscience. Nov. 12-16, 2022. *Nanosymposium talk*.

Mohan, U., Jacobs, J. Phase-specific modulation of endogenous low-frequency oscillations with direct electrical stimulation of the human cortex. BrainStim 2020 conference. May 19-20, 2020. *Online talk*.

Mohan, U. The effects of direct brain stimulation in humans depend on frequency and location. OptoDBS 2019. Campus Biotech Geneva. June 21, 2019. *Symposium talk*.

TRAVEL AWARDS AND FELLOWSHIPS:

Cognitive Neuroscience Society Postdoctoral Fellows Award (2026)
NINDS Fellows Award for Research Excellence Recipient (FARE 2026)
The Nakatani Foundation for Advancement of Measuring Technologies in Biomedical Engineering, Travel grant via Exchange Grant Program
NINDS Fellows Award for Research Excellence Recipient (FARE 2025)
Minnesota Neuromodulation Symposium, 2020
OptoDBS 2019. Campus Biotech Geneva. June, 2019.
IEEE Brain Control Interface Hackathon, Temple University, Second Place, 2016
Provost's Diversity Fellowship at Columbia University, 2015.
MIT Hacking Medicine, Grand Hack at MIT, Telehealth Track-Second place, 2015
HackMed NYC, Health solutions focused hackathon hosted by Cornell Tech and MIT, First place, 2014.
Johns Hopkins Business Plan Competition. Finalist, 2012.
Beyond Tradition Borders. National Undergraduate Global Health Design Competition hosted by Rice University. Awarded 3rd place, 2012
Siemens Competition in Math, Science, and Engineering Semifinalist, 2009.
Intel International Science and Engineering Fair attendee, 2008.

TEACHING:

Mount St. Mary's University — Emmitsburg, MD
Biostatistics, Adjunct Professor — Summer 2022,2023,2024, 2025, 2026

Columbia University: Department of Biomedical Engineering — New York, NY
Ethics for Biomedical Engineers, Graduate Teaching Assistant — Spring 2016
BME Senior Design I/II, Graduate Teaching Assistant — Fall 2016-Spring 2017
Electrophysiology of Memory and Navigation, Graduate Teaching Assistant — Spring 2018

Cornell University: Department of Biomedical Engineering — Ithaca, NY
Introduction to Biomedical Engineering, Graduate Teaching Assistant — January-May 2014

MENTORING:

Rahil Verma (06/2024-present). Summer research intern. Project: *The Characteristics of Cortical Traveling Waves in Human Attention and Memory*

David Zarrin (7/2024-present). Medical Research Scholars Program Fellow. Project: *Hippocampal Micro-Stimulation Modulates High-Frequency Ripple Oscillations in Humans*

Helen Qian (06/2023-02/2025). Summer research intern. Project: *Electrical microstimulation alters burst rate and sequence structure of neuronal spiking activity in the human cortex*.

Nuha Mohammed (06/2024-08/2024). Summer research intern. Project: *Using Human Intracranial EEG Recordings to Examine Ripple Characteristics during Cued Attention and Memory Encoding*

William Noll (06/2023-05/2024). Summer research intern. Project: *Effects of direct cortical stimulation on ripple characteristics in humans*

Molly Baumhauer (07/2022-05/2023). Postbaccalaureate Research Fellow. Project: *Neural dynamics in the human brain underlying successful memory formation and retrieval*

John Mutersbaugh (06/2022-08/2022). Summer research intern. Project: *Classifying Memory Processing States by using Convolutional Deep Learning*

Nicholas Lorusso (05/2021-03/2022). Master's thesis mentorship. Thesis: *Deep Brain Stimulation for Therapeutic Intervention*. Publication: *Jose Delgado: A controversial trailblazer in neuromodulation*.

Claire Padilla (05/2021-08/2021). Brain Research Apprenticeship in New York (BRAINYAC) program at Columbia University. Project: *There is No Statistically Significant Difference Between High and Low BAI Scores*.

Max Rachmuth (05/2021-08/2021). Brain Research Apprenticeship in New York (BRAINYAC) program at Columbia University. Projects: *Correlation Between Depression and Verbal Memory Performance*.

ACADEMIC SERVICE:

Founding organizer of **Wave Club Seminar**: a monthly seminar series dedicated to exploring neuronal traveling waves with leading experts in the field (2024-present)

Peer reviewing: Nature Neuroscience, eLife, Journal for Neuroscience Research (JNR), Clinical Neurophysiology Practice, Imaging Neuroscience

SOCIETY MEMBERSHIPS:

Society for Neuroscience (2016-present)
Organization for Human Brain Mapping (2024)
Cognitive Neuroscience Society (2015 ,2026)

COLLABORATORS:

Joshua Jacobs (University of Chicago)
Salman Qasim (Rutgers University)
Bard Ermentrout (University of Pittsburgh)

Michael Kahana (University of Pennsylvania, Nia Therapeutics Inc.)
Bradley Lega (Neurosurgery, University of Texas, Southwestern Medical Center)
Rachel J. Smith (University of Alabama at Birmingham)
Adam Goodman (University of Alabama at Birmingham)

MEDIA COVERAGE (selected):

Press related to "The direction of theta and alpha travelling waves modulates human memory processing" in Nature Human Behavior (2024): Science Alerts, Earth, Discover Magazine

Press related to "Neuronal activity in the human amygdala and hippocampus enhances emotional memory encoding." in Nature Human Behavior (2024): NewScientist

POSTER PRESENTATIONS (selected):

Mohan, U.R., Zarrin, D., Fruchet, O.E., Sundby, K.K., Xie, W., Gau, Y.A., Serlin, Y., Inati, S., Zaghloul, K.A. Direct electrical microstimulaiton evokes high-frequency ripple oscillations in humans. Society for Neuroscience. Nov. 15-19, 2025.

Zarrin, D., **Mohan, U.R**., Fruchet, O.E., Sundby, K.K., Chapeton, J.I., Xie, W., Inati, S., Zaghloul, K.A. Hippocampal Micro-Stimulation Modulates High-Frequency Ripple Oscillations in Humans. Society for Neuroscience. Oct. 5-9, 2024.

Favila S.E., **Mohan U.R.**, Jacobs J. Coordinated traveling waves between the MTL and neocortex support memory encoding. Society for Neuroscience. Oct. 5-9, 2024.

Zarrin, D., **Mohan, U.R**., Sundby, K.K., Zaghloul, K.A. Hippocampal Micro-Stimulation Modulates High-Frequency Ripple Oscillations in Humans. Congress of Neurological Surgeons. Sept 28-Oct 02, 2024.

- Awarded best functional clinical research poster

Mohan, U.R., Fruchet O.E., Wittig, J.H., Inati, S.K., Zaghloul, K.A. Modeling and predicting neural responses to multisite direct electrical brain stimulation in humans. 2024 Organization for Human Brain Mapping. June 23-June 27, 2024.

Mohan, U.R., Fruchet O.E., Wittig, J.H., Inati, S.K., Zaghloul, K.A. Modeling and predicting neural responses to multisite direct electrical brain stimulation in humans. 2024 Context and Episodic Memory Symposium. May 30-May 31, 2024.

Mohan, U.R., Fruchet O.E., Wittig, J.H., Inati, S.K., Zaghloul, K.A. Modeling and predicting neural responses to multisite direct electrical brain stimulation in humans. Society for Neuroscience. Nov. 11-15, 2023.

Mohan, U.R., Zhang, H., Ermentrout, B., Jacobs, J. Direct cortical stimulation modulates propagation directions of traveling wave in humans**.** 2023 Context and Episodic Memory Symposium. May 31-June 1, 2023.

Baumhauer, M., **Mohan, U.R.**, Chapeton, J., Weizhen, X., Zaghloul, K.A. Neural dynamics underlying successful memory formation and retrieval. Society for Neuroscience. Nov. 12-16, 2022.

Hermiller, M., **Mohan, U.R.**, Jacobs, J. Frequency-specific response to intracranial theta-burst stimulation in the human hippocampal-cortical network. Society for Neuroscience. Nov. 12-16, 2022.

Mohan, U., Wittig, J. H., Inati, S. K., Zaghoul, K.A**.** Characterizing single-neuron responses to direct electrical brains stimulation in humans**.** 2022 Human Single Neuron Meeting. Nov 10-11, 2022.

Mohan, U., Zhang, H., Jacobs, J**.** Direct cortical stimulation modulates propagation directions of traveling wave in humans. 2022 Joint meeting of Neuroergonomics & NYC Neuromodulation. July 28-Aug 1, 2022.

Mohan, U., Zhang, H., Jacobs, J. The direction and timing of cortical traveling waves modulate human memory processes. Society for Neuroscience. *Virtual.* Nov 8-11, 2021.

Mohan, U., Zhang, H., Jacobs, J. The direction and timing of cortical traveling waves modulate human memory processes. Context and Episodic Memory Symposium. Aug 16-17, 2021.

Mohan, U., Jacobs, J. Direct electrical stimulation of the human cortex modulates narrowband low frequency oscillations. Society for Neuroscience. Oct. 21, 2019.

Mohan, U., Sperling, M., Sharan, A., Worrell, G., Berry, B., Lega, B., Jobst, B., Davis, K., Gross, R., Sheth, S., Das, S., Stein, J., Gorniak, R., Rizzuto, D., Kahana, M. Jacobs, J. High and low frequency stimulation have inverse effects on neuronal activity. Society for Neuroscience. Nov. 3, 2018.

Mohan, U., Watrous, A. , Sperling, M. , Sharan, A., Worrell, G., Berry, B., Lega, B., Jobst, B., Davis, K., Gross, R., Sheth, S., Das, S., Stein, J., Gorniak, R., Rizzuto, D., Kahana, M. Jacobs, J. Changes in high frequency activity depend on location of electrical stimulation in the human cortex. 2018 NYC Neuromodulation Conference & NANS Summer Series. Aug. 23, 2018.

Mohan, U., Watrous, A. , Sperling, M. , Sharan, A., Worrell, G., Berry, B., Lega, B., Jobst, B., Davis, K., Gross, R., Sheth, S., Das, S., Stein, J., Gorniak, R., Rizzuto, D., Kahana, M. Jacobs, J. Changes in high frequency activity depend on location of electrical stimulation in the human cortex. Society for Neuroscience. Nov. 12, 2017.

Mohan, U., Miller, J., Watrous, A., Lee, S.A. , Sperling, M. , Sharan, A., Worrell, G., Berry, B., Lega, B., Jobst, B., Davis, K., Gross, R., Sheth, S., Das, S., Stein, J., Gorniak, R., Rizzuto, D., Jacobs, J. Direct brain recordings reveal patterns of theta and alpha oscillations related to spatial navigation and memory. Society for Neuroscience. Nov. 14, 2016.

Mohan, U., Sukhatme, S., Venkatesh, P. A Wearable Device for Measuring Hydration and Body Fat. Electrical and Computer Engineering Day. May 7, 2014.

Mohan, U., Wei, T., Cook, A. Mathematical Modeling and Simulation of Anesthetic Drug Delivery Devices: Implications Affecting Closed-Loop Control Technologies. BMES Research Day-Johns Hopkins University. Mar. 2013.

Gurbani, S., **Mohan, U.,** Wei, T., Cook, A. Mathematical Modeling and Simulation of Anesthetic Drug Delivery Devices: Implications Affecting Closed-Loop Control Technologies. Annual BMES conference. Oct. 26, 2012.

Rajan, V., Dasgupta, R., Ukaigwe, J., Powers, R.J., **Mohan, U**., Jiam, H., Rao, P., Mandel, J., Kondragunta, R., Kang, M. G6PDesign: A Point of Care (POC) Diagnostic to Detect Glucose-6-Phosphate Dehydrogenase (G6PD) Enzyme Deficiency to Improve Treatment for Malaria. Johns Hopkins University Center for Bioinnovation and Design Design Day. May 7, 2012.

Rajan, V., Dasgupta, R., Ukaigwe, J., Powers, R.J., **Mohan, U**., Jiam, H., Rao, P., Mandel, J., Kondragunta, R., Kang, M. G6PDesign: A Point of Care (POC) Diagnostic to Detect Glucose-6-Phosphate Dehydrogenase (G6PD) Enzyme Deficiency to Improve Treatment for Malaria. Johns Hopkins Public Health Conference. Apr. 27, 2012.

Rajan, V., Dasgupta, R., Ukaigwe, J., Powers, R.J.**, Mohan, U**., Jiam, H., Rao, P., Mandel, J., Kondragunta, R., Kang, M. G6PDesign: A Point of Care (POC) Diagnostic to Detect Glucose-6-Phosphate Dehydrogenase (G6PD) Enzyme Deficiency to Improve Treatment for Malaria. Johns Hopkins Business Plan Competition. Finalists. Apr. 27, 2012.

Rajan, V., Dasgupta, R., Ukaigwe, J., Powers, R.J., **Mohan, U**., Jiam, H., Rao, P., Mandel, J., Kondragunta, R., Kang, M. G6PDesign: A Point of Care (POC) Diagnostic to Detect Glucose-6-Phosphate Dehydrogenase (G6PD) Enzyme Deficiency to Improve Treatment for Malaria. Beyond Traditional Borders National Undergraduate Global Health. Design Competition hosted by Rice University. Mar. 30, 2012.

Mohan, U. All, A. Repair of Axonal Injury to Simple Leech Central Nervous System Model. Johns Hopkins University. Biomedical Engineering Research Day. Dec. 2, 2010.

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